Math 208 - Tutorial 5

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Example 3.1 *Goodwin Manufacturing Company*

Goodwin Manufacturing Company is planning next week's shipments from its three manufacturing plants to its four distribution warehouses and is seeking a minimum-cost shipping schedule. Each plant has a potential capacity, expressed in cartons of product, and each warehouse has a demand requirement for the week that must be met. There are 12 possible shipment routes, and for every plant–warehouse combination, the unit shipping cost is known. The following table provides the given information.

(From) Plant	**Atlanta**	**Boston**	**Chicago**	**Denver**	**Capacity**
Minneapolis	\$0.60	\$0.56	\$0.22	\$0.40	10,000
Pittsburgh	\$0.36	\$0.30	\$0.28	\$0.58	15,000
Tucson	\$0.65	\$0.68	\$0.55	\$0.42	15,000
Requirement	8,000	10,000	12,000	9,000	

Table 1: Shipping Costs and Capacities

4.1 Goodwin Manufacturing Model Revisited

Revisit the transportation model of this chapter and its optimal solution.

(a) Suppose that the unit cost for the Tucson–Atlanta route is open for negotiations. Perform a sensitivity analysis on this cost for a range of \$0.60–\$0.65 per carton. Construct a table showing the value of the optimal cost and the quantities shipped to Atlanta from each of the three plants.

(b) Suppose instead that expansion of Minneapolis plant is under consideration. Perform a sensitivity analysis on this capacity for a range of 10,000–30,000 cartons. Construct a table showing the value of the optimal cost in this range and produce a chart showing the results graphically.

Solution

Part(a):

Using the Sensitivity Toolkit on the Tucson-Atlanta route with parameter range 0.6 - 0.65, increment by 0.01, we obtain the following table:
The green highlighted in the table is the insensitive zone in the parameter range

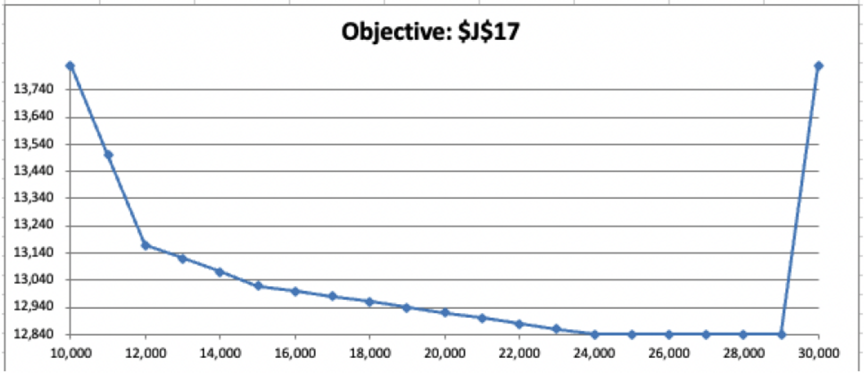
PA Cost	Objective	MA	PA	TA
0.60	13,620	0	3000	5000
0.61	13,670	0	3000	5000
0.62	13,720	0	3000	5000
0.63	13,770	0	3000	5000
0.64	13,800	0	5000	3000
0.65	13,830	0	5000	3000

Part(b):

We now use the Sensitivity Toolkit on the capacity of the Minneapolis plant with parameter range from 10,000 to 30,000, increment by 1000, we obtain the following table: Looking at

Minneapolis Capacity	Optimal Cost	MA	MB	MC	MD
10,000	13,830	0	0	10000	0
11,000	13,500	0	0	11000	0
12,000	13,170	0	0	12000	0
13,000	13,120	1000	0	12000	0
14,000	13,070	2000	0	12000	0
15,000	13,020	3000	0	12000	0
16,000	13,000	3000	0	12000	1000
17,000	12,980	3000	0	12000	2000
18,000	12,960	3000	0	12000	3000
19,000	12,940	3000	0	12000	4000
20,000	12,920	3000	0	12000	5000
21,000	12,900	3000	0	12000	6000
22,000	12,880	3000	0	12000	7000
23,000	12,860	3000	0	12000	8000
24,000	12,840	3000	0	12000	9000
25,000	12,840	3000	0	12000	9000
26,000	12,840	3000	0	12000	9000
27,000	12,840	3000	0	12000	9000
28,000	12,840	3000	0	12000	9000
29,000	12,840	3000	0	12000	9000
30,000	13,830	0	0	10000	0

the price of the graph, we can see that increasing the plant capacity beyond 24,000 units, does not improve on our objective function and it actually increases when we increase the plant capacity to 30,000.



4.4 Make or Buy(Tutorial 2-Second question)

A sudden increase in the demand for smoke detectors has left Acme Alarms with insufficient capacity to meet demand. The company has seen monthly demand for its electronic and battery-operated detectors rise to 20,000 and 10,000, respectively, and Acme wishes to continue meeting demand. Acme's production process involves three departments: fabrication, assembly, and shipping. The relevant quantitative data on production and prices are summarized below.

Department	Monthly Hours Available	Hours/Unit (Electronic)	Hours/Unit (Battery)
Fabrication	2000	0.15	0.10
Assembly	4200	0.20	0.20
Shipping	2500	0.10	0.15

Cost/Price	Electronic Detector	Battery Detector
Variable Cost/Unit ($)	18.80	16.00
Retail Price/Unit ($)	29.50	28.00

The company also has the option of obtaining additional units from a subcontractor, who has offered to supply up to 20,000 units per month in any combination of electronic and battery-operated models, at a charge of $21.50 per unit. For this price, the subcontractor will test and ship its models directly to the retailers without using Acme's production process.

(a) Determine how the manufacturer should allocate its in-house capacity and how it should utilize the subcontractor. What are the maximum profit and the corresponding make/buy levels? (This is a planning model; fractional decisions are acceptable.)

(b) Investigate the solution for Shipping capacities between 1200 and 2400 hours. Draw a graph showing how the optimal quantities change over this range.

Solution:

Part(a):

Generating a sensitivity report, we get the following result: Here we can see that the shadow

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
B10	Unit Electronic (make)	6666.666667	0	10.7	5.55	2.7
C10	Unit Battery (make)	10000	0	12	1E+30	3.7
D10	Unit Electronic (buy)	13333.33333	0	8	2.7	5.55
E10	Unit Battery (buy)	0	-3.7	6.5	3.7	1E+30

Constraints

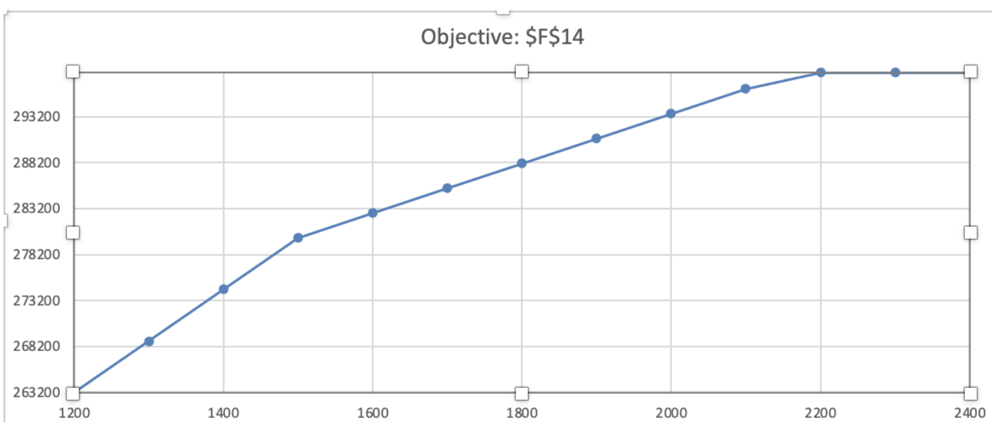
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
F18	Fabrication Hours Total	2000	18	2000	500	1000
F19	Assembly Hours Total	3333.333333	0	4200	1E+30	866.6666667
F20	Shipping Hours Total	2166.666667	0	2500	1E+30	333.3333333
F21	Subcontractor Purchase Total	13333.33333	0	20000	1E+30	6666.666667
F22	Demand (Electronics) Total	20000	8	20000	6666.666667	13333.33333
F23	Demand (Battery) Total	10000	10.2	10000	4000	10000

We can increase fabrication hours by 500, which will increase by 18*500 = 9000 dollars in profit
Assembly, Shiping, Subcontractor all have a shadow price of 0, suggesting that increasing the constraint will not increase profits

price for assembly, shipping, and subcontractor constraints are all 0, suggesting that if we increase the rhs of the constraint, we will not improve on our objective function. Fabrication total hours have a shadow price of 18 and an allowable increase of 500 hours in the insensitivity zone. This suggests that we should increase the hours by 500 to obtain \$9000 in additional profits.

Part(b)

We run the solver sensitivity with a lower bound of 1200, an upper bound of 2400, and an increment of 100 for the shipping capacity. We obtain the following graph: We can see a



steep increase from 1200 hours to 1500 hours, with it tapering off, and from 2200 hours to 2400 hours, the objective function does not improve.